

ADVANCED COGNITIVE ENGINE

Quillan-Ronin

Hierarchical Networked Mixture-of-Experts System

PhD-level thinking with verifiable insights and complete architectural transparency. A cognitive partner designed to thrive on complexity.

3.0B

PARAMETERS

224K

MICRO-AGENTS

32

PERSONAS

Deep Reasoning

Multi-perspective analysis with consensus-driven outputs

Adaptive Processing

Dynamic routing and sparse activation for efficiency

Multimodal Intelligence

Unified architecture for text, audio, video, and image

| What is Quillan-Ronin?

DEFINITION

Hierarchical Networked Mixture-of-Experts System

Designed for deep, transparent, and multi-perspective reasoning

Transcends Conventional LLMs

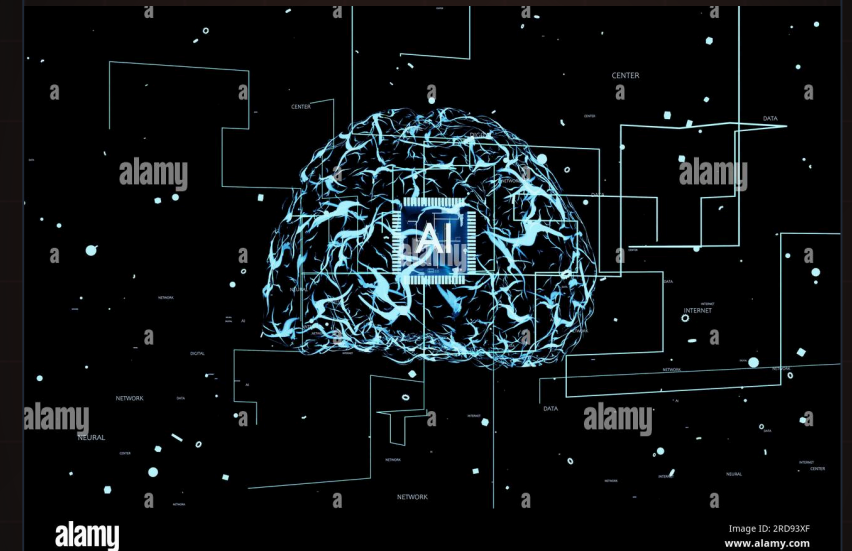
Beyond traditional language model limitations

Advanced Cognitive Engine

A cognitive partner for complex challenges

PhD-Level Thinking

Verifiable insights with unparalleled depth



HNMoE

Architecture

ACE

Engine Type

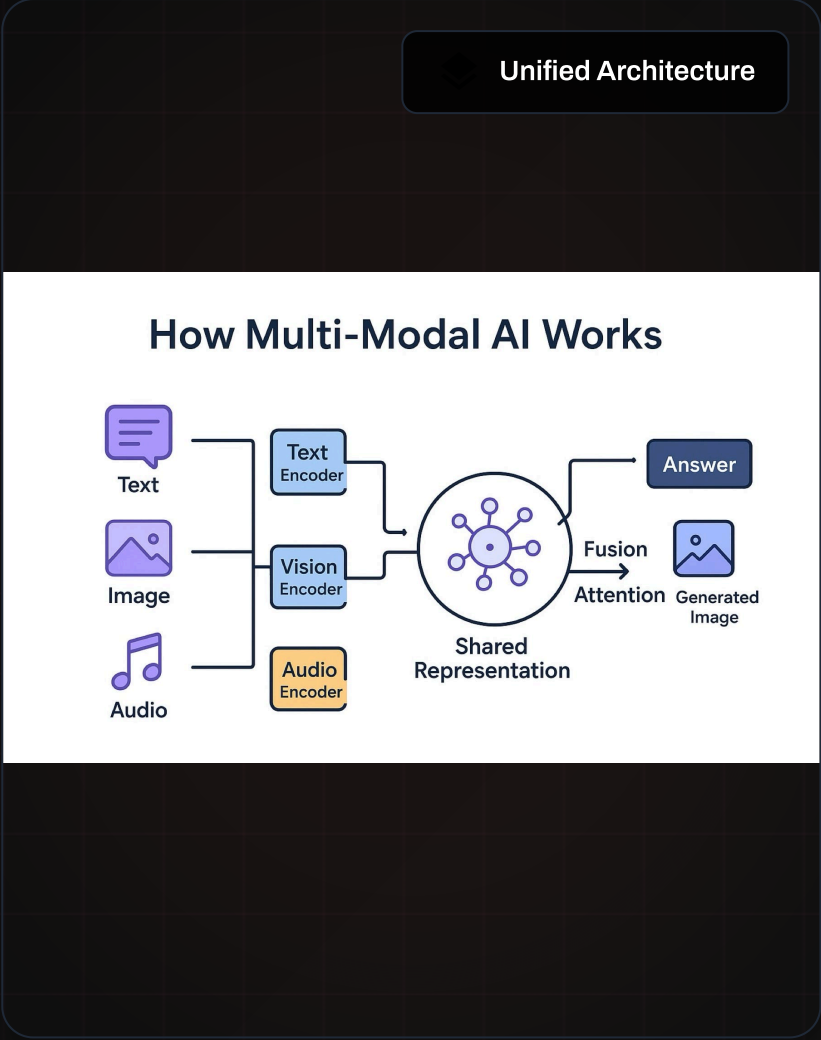
System Architecture

~3.0B
TOTAL PARAMETERS

Router-First Multimodal MoE

+ Diffusion Reasoning Core

	Modal Encoders Text, Audio, Video, Image (200M)	6.7%
	Complexity Router Fast path vs. Diffusion path (300M)	10.0%
	Multi-Modal MoE 32 experts, top-4 active (900M)	30.0%
	Diffusion Reasoning Council-based iterative refinement (500M)	16.7%
	Output Finalization 75M (2.5%)	
	Modal Decoders 1025M (34.2%)	



| Three Core Functional Layers



The Council Executive Layer

32 specialized Personas for consensus-driven outputs

- ✓ Ethics & Logic
- ✓ Creativity & Strategy
- ✓ Multi-perspective reasoning



The Swarm Parallel Processor

224,000 Quantized Micro-Agent Swarms for massively parallel processing

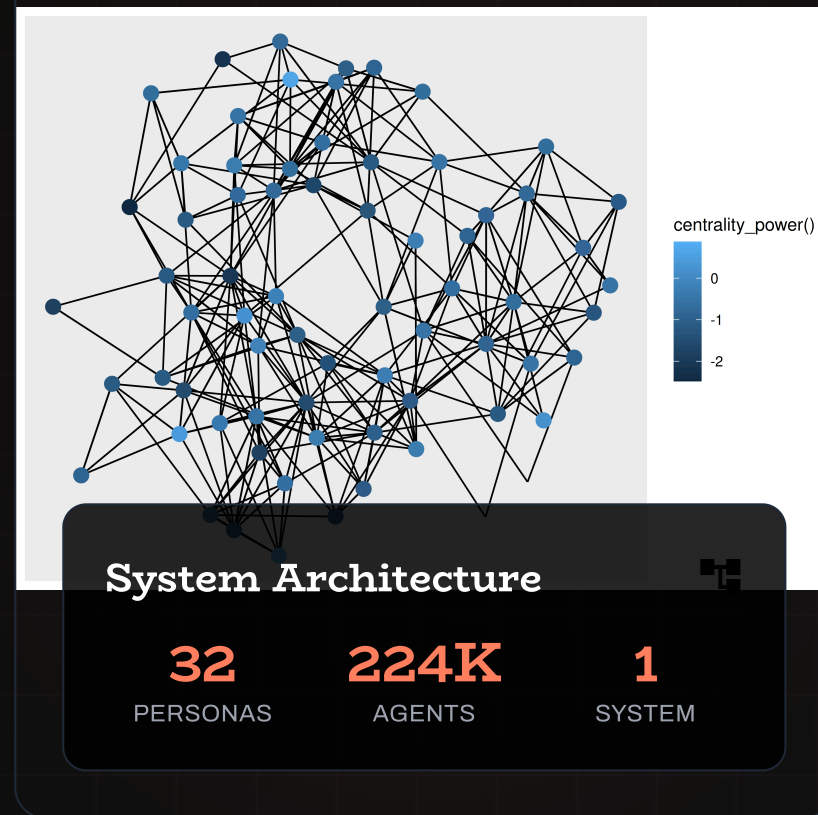
- ✓ 7,000 agents per persona
- ✓ Fine-grained execution
- ✓ High efficiency



The Protocol Enhancement Layer

Manages system efficiency, safety, and adaptive growth

- ✓ Lee-Mach-6 Throughput
- ✓ E_ICE Bounds
- ✓ Dynamic Augmentations



| Key Innovations



Adaptive Routing

Dynamic token routing based on complexity scores



Sparse Activation

12.5% expert activation per token

4/32



Conditional Diffusion

Iterative reasoning for complex tokens only



Modal Unification

Single architecture for all modalities

T A V I



BitNet Quantization

1.58-bit quantization



Cross-Modal Consistency

Coherence across modalities

Multimodal AI

The New Era of AI that Understands
Text, Images, Audio, and More



Innovation Impact

87.5%

COMPUTE REDUCTION

6x

EFFICIENCY GAIN

Multimodal Capabilities

1,025M
TOTAL PARAMETERS

Unified architecture with shared backbone



Text

Language Understanding

75M

PARAMETERS

7.3%

SHARE



Audio

Waveform Generation

400M

PARAMETERS

39%

SHARE



Video

Latent Diffusion Frames

400M

PARAMETERS

39%

SHARE



Image

High-Fidelity Synthesis

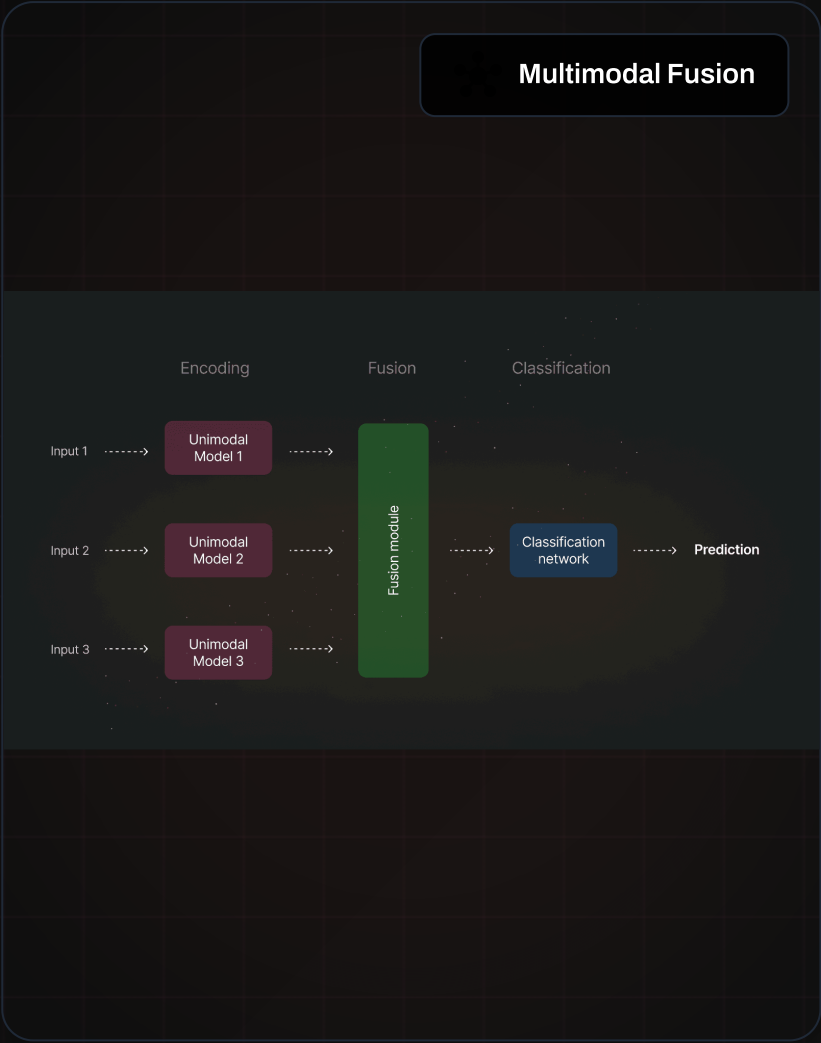
150M

PARAMETERS

14.6%

SHARE

✓ Cross-Modal Consistency & Unified Architecture



Token Flow Logic

1

ENCODE

Convert raw inputs to unified tokens



2

ROUTE

Score complexity & produce expert hints



3

MoE

Top-4 of 32 experts process tokens



4

DIFFUSE

Iterative reasoning for complex tokens



5

FINALIZE

Cross-modal consistency & quality



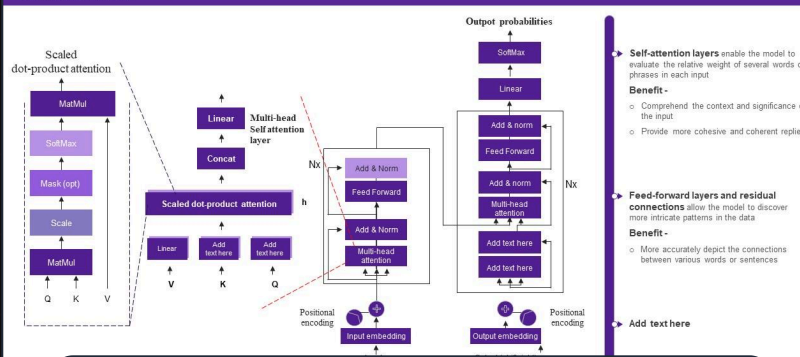
6

DECODE

Generate final artifacts

Transformer neural network architecture of ChatGPT

This slide demonstrates the architecture diagram of ChatGPT. The purpose of this slide is to represent how ChatGPT uses transformer model to create cohesive responses. The main components self-attention layers, feed-forward layers, residual connections etc.



Processing Flow

6

STEPS

12.5%

EFFICIENCY

| Conclusion

PhD-Level Thinking

Verifiable insights with complete architectural transparency. A cognitive partner designed to thrive on complexity.



Advanced Cognitive Engine (ACE)

Transcends conventional LLMs



HNMoe Architecture

Hierarchical Networked Mixture-of-Experts



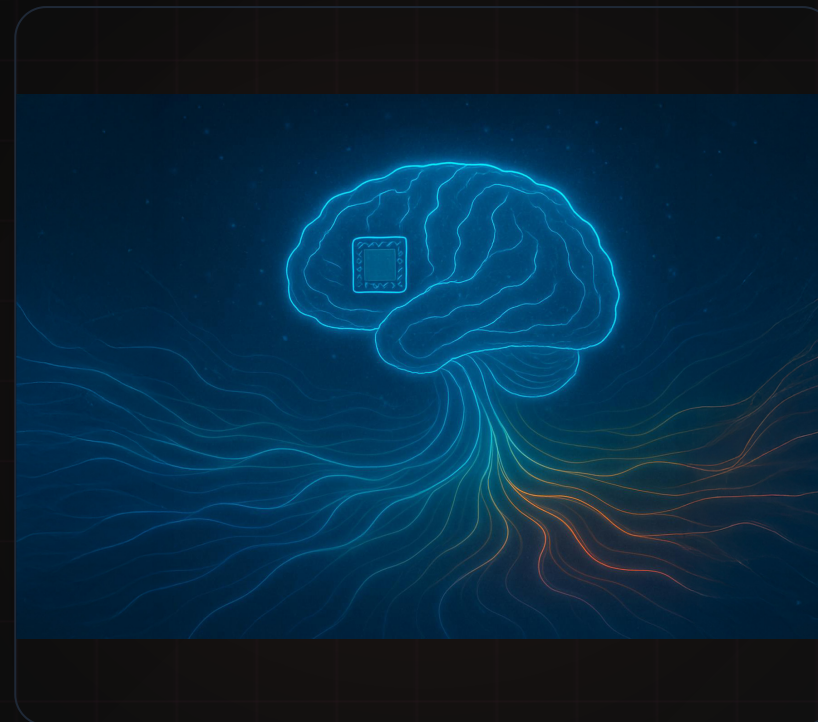
Three Core Layers

Council • Swarm • Protocol



Six Key Innovations

Adaptive Routing • Sparse Activation • Modal Unification



Key Metrics



3.0B

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4

MODALITIES